

8.1 Warm-Up Exercises

In the following exercises, all coordinates and components are given with respect to a right-oriented orthonormal frame $\mathcal{K}$.

8.1. The vertices of a triangle are $A(1, 1)$, $B(4, 1)$ and $C(2, 3)$. Determine the image of the triangle ABC under a rotation by 90° around C followed by an orthogonal reflection relative to the line AB .

8.2. Verify that the matrices

$$A = \frac{1}{3} \begin{bmatrix} -1 & 2 & -2 \\ -2 & -2 & -1 \\ -2 & 1 & 2 \end{bmatrix} \quad \text{and} \quad B = \frac{1}{11} \begin{bmatrix} -9 & -2 & 6 \\ 6 & -6 & 7 \\ 2 & 9 & 6 \end{bmatrix}$$

belong to $SO(3)$. Moreover, in each case determine the axis of rotation and the rotation angle.

8.3. Using Euler-Rodrigues formula, deduce the known matrix forms of rotations around the coordinate axes.

8.4. Using rotations around the coordinate axes, give a parametrization of a cylinder with axis $\mathbb{R}\mathbf{v}$ and diameter $\sqrt{2}$.

$$\mathbf{v} = (0, 1, 1)$$

8.1. The vertices of a triangle are $A(1,1)$, $B(4,1)$ and $C(2,3)$. Determine the image of the triangle ABC under a rotation by 90° around C followed by an orthogonal reflection relative to the line AB .

$$[Rot_{90^\circ}] = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

$$Rot_{C, 90^\circ} = T_{\vec{OC}} \cdot Rot_{90^\circ} \cdot T_{-\vec{OC}}$$

$$\begin{aligned} Rot_{C, 90^\circ} \begin{bmatrix} x \\ y \end{bmatrix} &= \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \left(\begin{bmatrix} x \\ y \end{bmatrix} - \begin{bmatrix} 2 \\ 3 \end{bmatrix} \right) + \begin{bmatrix} 2 \\ 3 \end{bmatrix} \\ &= \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x-2 \\ y-3 \end{bmatrix} + \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 3-y \\ x-2 \end{bmatrix} + \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 5-y \\ x+1 \end{bmatrix} \end{aligned}$$

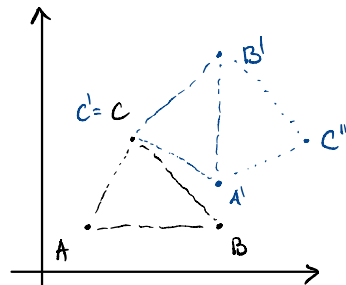
in particular $Rot_{C, 90^\circ}(A) = \begin{bmatrix} 4 \\ 2 \end{bmatrix} = A'$ $Rot_{C, 90^\circ}(B) = \begin{bmatrix} 4 \\ 5 \end{bmatrix} = B'$

So the line AB moves to $A'B'$ and a reflection in $A'B'$ can be read-off from the picture

$$Ref_{A'B'}^\perp \begin{bmatrix} x \\ y \end{bmatrix} = T_{(-4,0)} \cdot Ref_{x=4}^\perp \cdot T_{(4,0)}$$

$$= \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \left(\begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} 4 \\ 0 \end{bmatrix} \right) + \begin{bmatrix} 4 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x+4 \\ y \end{bmatrix} + \begin{bmatrix} 4 \\ 0 \end{bmatrix} = \begin{bmatrix} 4-x \\ y \end{bmatrix} + \begin{bmatrix} 4 \\ 0 \end{bmatrix} = \begin{bmatrix} 8-x \\ y \end{bmatrix}$$

in particular $Ref_{A'B'}^\perp(A') = A'$, $Ref_{A'B'}^\perp(B') = B'$, $Ref_{A'B'}^\perp(C') = \begin{bmatrix} 6 \\ 3 \end{bmatrix} = C''$



8.2. Verify that the matrices

$$A = \frac{1}{3} \begin{bmatrix} -1 & 2 & -2 \\ -2 & -2 & -1 \\ -2 & 1 & 2 \end{bmatrix} \quad \text{and} \quad B = \frac{1}{11} \begin{bmatrix} -9 & -2 & 6 \\ 6 & -6 & 7 \\ 2 & 9 & 6 \end{bmatrix}$$

belong to $SO(3)$. Moreover, in each case determine the axis of rotation and the rotation angle.

• $A \in SO(3) \Leftrightarrow AA^T = I_3 \text{ \& } \det A = 1$

$$A^T = \frac{1}{3} \begin{bmatrix} -1 & -2 & -2 \\ 2 & -2 & 1 \\ -2 & -1 & 2 \end{bmatrix}$$

$$A \cdot A^T = \frac{1}{3} \begin{bmatrix} -1 & 2 & -2 \\ -2 & -2 & -1 \\ -2 & 1 & 2 \end{bmatrix} \cdot \frac{1}{3} \begin{bmatrix} -1 & -2 & -2 \\ 2 & -2 & 1 \\ -2 & -1 & 2 \end{bmatrix} = \frac{1}{9} \begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I_3$$

One checks that $\det A = 1$ so, yeah, $A \in SO(3)$

• the axis of rotation is the line passing through the origin in the direction of the eigenvectors for the eigenvalue 1

this is obtained by solving $A \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \Leftrightarrow \frac{1}{3} \begin{bmatrix} -1 & 2 & -2 \\ -2 & -2 & -1 \\ -2 & 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$

$$\Leftrightarrow \begin{bmatrix} -4 & 2 & -2 \\ -2 & -5 & -1 \\ -2 & 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -4 & 2 & -2 \\ -2 & -5 & -1 \\ -2 & 1 & -1 \end{bmatrix} \sim \begin{bmatrix} 2 & -1 & 1 \\ -2 & -5 & -1 \\ -2 & 1 & -1 \end{bmatrix} \sim \begin{bmatrix} 2 & -1 & 1 \\ 0 & -6 & 0 \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow y = 0 \quad \begin{cases} \Rightarrow 2x + z = 0 \\ \Rightarrow z = -2x \end{cases}$$

$\Rightarrow$ eigenspace for $\lambda = 1$ is $V_1 = \{ (t, 0, -2t) : t \in \mathbb{R} \}$ this is a line passing through the origin, it is the rotation axis
the eigenvectors are the non-zero vectors in V_1

• the angle of rotation θ is determined by

$$\text{tr}(A) = 1 + 2 \cos \theta \quad (\Rightarrow) \quad \cos \theta = \frac{\text{tr} A - 1}{2} = \frac{-\frac{1}{3} - 1}{2} = -\frac{2}{3}$$

$$\text{so } \theta = \arccos\left(-\frac{2}{3}\right)$$

the calculation for B is similar.

8.3. Using Euler-Rodrigues formula, deduce the known matrix forms of rotations around the coordinate axes.

$$\begin{aligned} \text{Rot}_{v,\theta}(x) &= \cos \theta x + \sin \theta (v \times x) + (1 - \cos \theta) \underbrace{\langle v, x \rangle v}_{\substack{= \\ v \cdot \langle v, x \rangle \\ = \\ v \cdot (v^T \cdot x) \\ = \\ v \otimes v}} \quad \text{for axis } Rv \\ &= \left[\cos \theta I_n + \sin \theta [v \times \cdot] + (1 - \cos \theta) v \otimes v \right] \cdot x \end{aligned}$$

for a rotation around Ox we choose $v = i$

$$[i \times \cdot] = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} \quad i \otimes i = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \cdot [1 \ 0 \ 0] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{aligned} \Rightarrow \text{Rot}_{i,\theta}(x) &= \left(\begin{bmatrix} \cos \theta & 0 & 0 \\ 0 & \cos \theta & 0 \\ 0 & 0 & \cos \theta \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -\sin \theta \\ 0 & \sin \theta & 0 \end{bmatrix} + \begin{bmatrix} 1-\cos \theta & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \right) x \\ &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix} x \end{aligned}$$

Similar for Oy and Oz

8.4. Using rotations around the coordinate axes, give a parametrization of a cylinder with axis $\mathbb{R}\mathbf{v}$ and diameter $\sqrt{2}$.

$$\mathbf{v} = (0, 1, 1)$$

The cylinder contains the line

$$l: \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

and has axis $l': \begin{bmatrix} x \\ y \\ z \end{bmatrix} = t \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$



We may construct a cylinder $\tilde{\mathcal{C}}$ of diam. $\sqrt{2}$ and axis Oz and then rotate it in the right position

$$\tilde{\mathcal{C}} \ni \tilde{l}: \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ \sqrt{2}/2 \\ 0 \end{bmatrix} + t \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\tilde{\mathcal{C}} = \text{Rot}_{K,0}(\tilde{l}) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ \sqrt{2}/2 \\ t \end{bmatrix} = \begin{bmatrix} -\frac{\sqrt{2}}{2} \sin\theta \\ \frac{\sqrt{2}}{2} \cos\theta \\ t \end{bmatrix}$$

$$\Rightarrow \mathcal{C} = \text{Rot}_{i,-45^\circ}(\tilde{\mathcal{C}}) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \sqrt{2}/2 & \sqrt{2}/2 \\ 0 & -\sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix} \begin{bmatrix} -\frac{\sqrt{2}}{2} \sin\theta \\ \frac{\sqrt{2}}{2} \cos\theta \\ t \end{bmatrix} = \dots$$